

Wikipedia task (N = 101): pre-registered analysis null, reading-only streaks show a robust signal

Pre-registered analysis

PC1 of three switch counts (time, topic, typing)

d = +0.04, p = 0.83

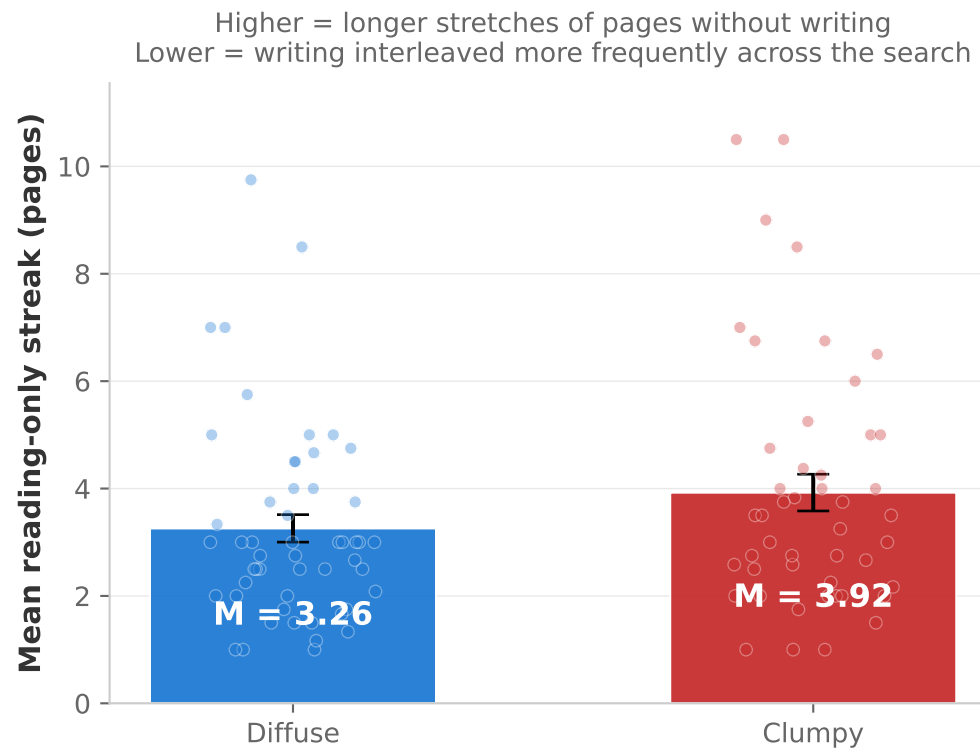
no condition difference

Exploratory finding

Mean reading-only streak

d = -0.44, p = 0.038

Clumpy > Diffuse



The manipulation does not move the pre-registered switch counts,
but it does change HOW writing is distributed across pages:

In Clumpy, the average reading-only stretch is longer - writing concentrates into bursts with longer non-writing gaps.

In Diffuse, writing is interleaved throughout the search, keeping reading-only stretches short.

The 5 writing-pattern measures, plus two robustness checks

(A) defines the measures used throughout. (B) and (C) re-analyze them from different angles.

(A) The 5 writing-pattern measures

used throughout panels B and C

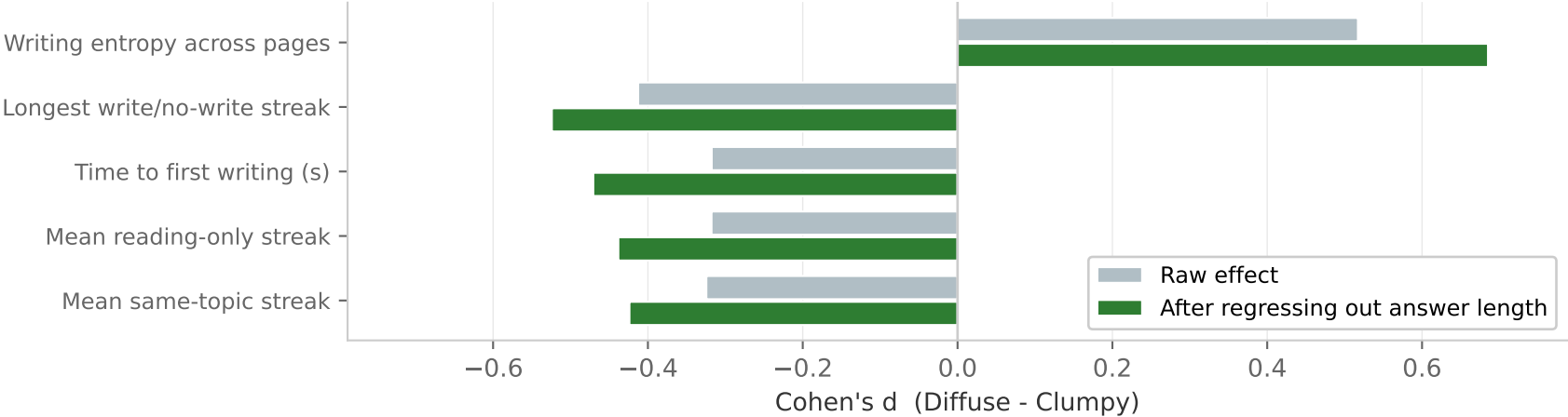
What they capture: each measure captures the W/N sequence from a different angle.

Writing entropy	- balance between W and N pages across the question
Longest write/no-write streak	- longest run of a single label (W or N)
Time to first writing (s)	- latency until the first writing on any page
Mean reading-only streak	- average length of consecutive no-write (N) pages
Mean same-topic streak	- average length of consecutive pages on the same LDA topic

(B) Does the effect survive controlling for output volume?

YES - it strengthens

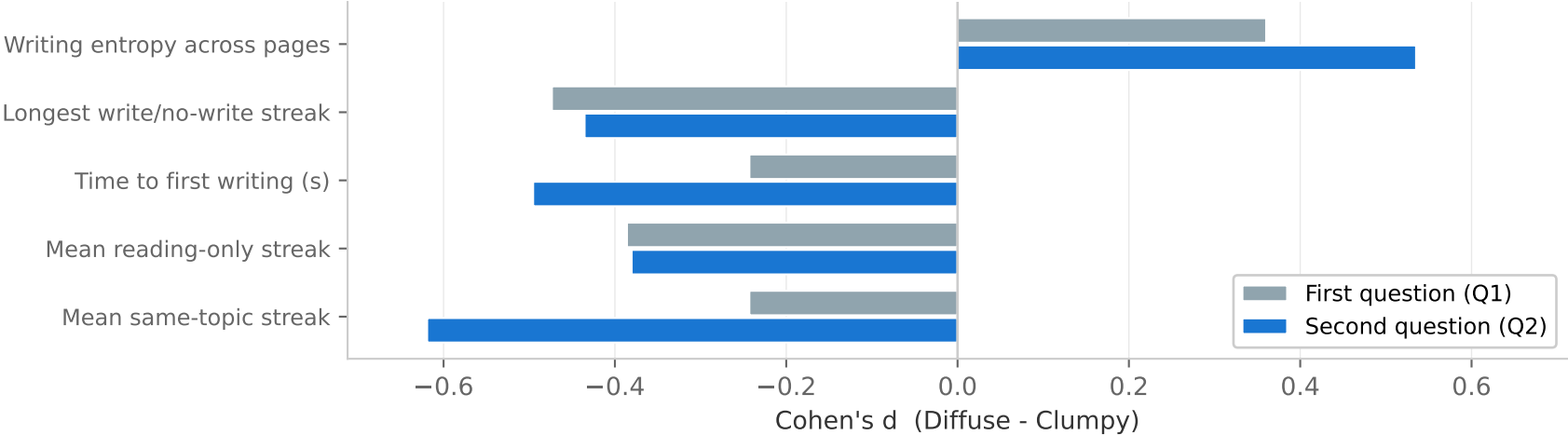
What the bars show: effect size for each measure, before vs after partialling out final answer length in a regression.



(C) Does the effect decay or grow across questions?

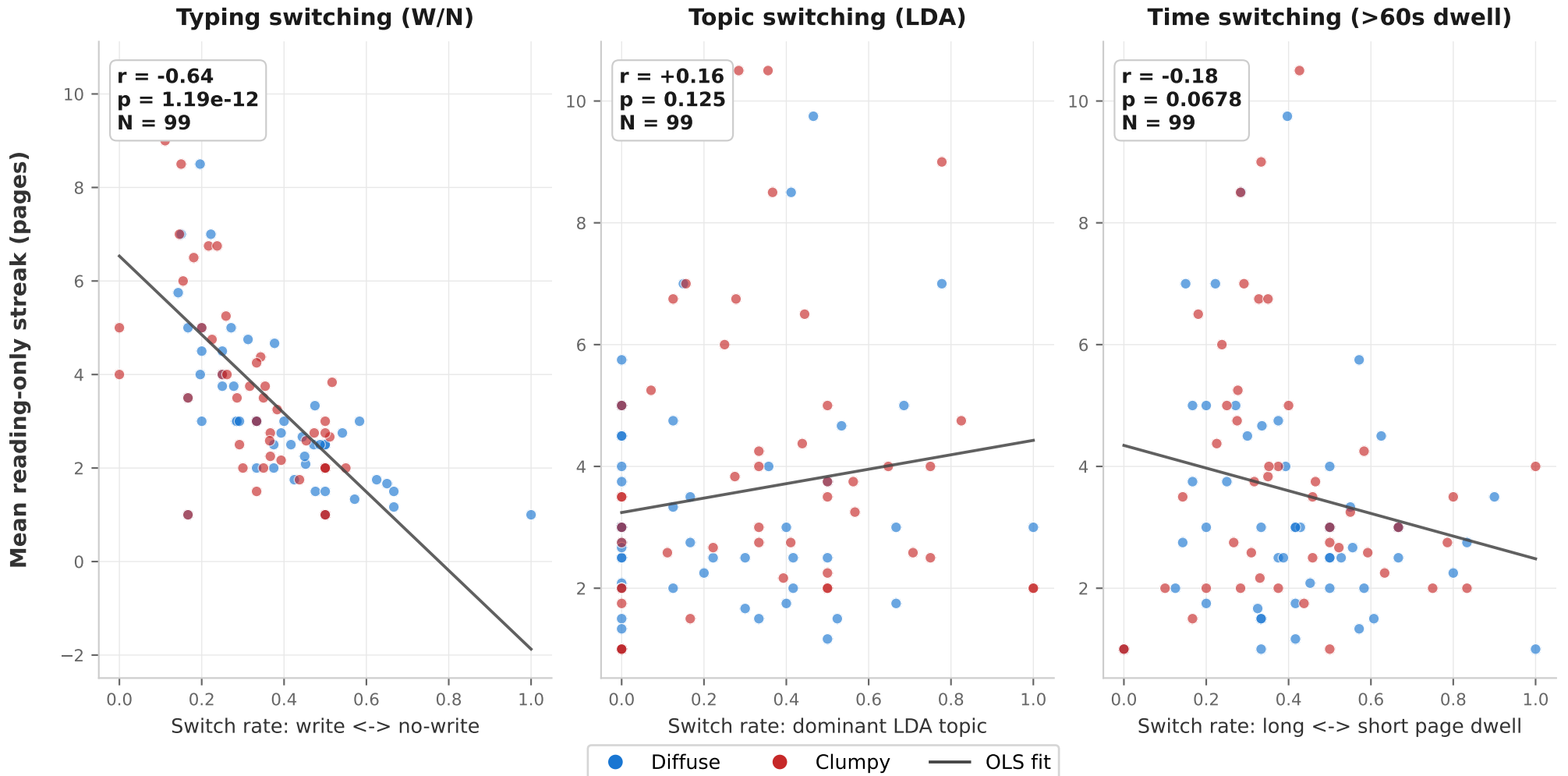
GROWS on Q2

What the bars show: effect size on participants' first vs second question (question order was randomised across participants).



How the Mean reading-only streak relates to the three original switch measures

Per participant, averaged across both questions (N = 99 of 101; 2 wrote on every page, leaving no reading-only streak)



Mean reading-only streak is mathematically tied to typing switching ($r = -0.64$) - they summarise the same W/N sequence.

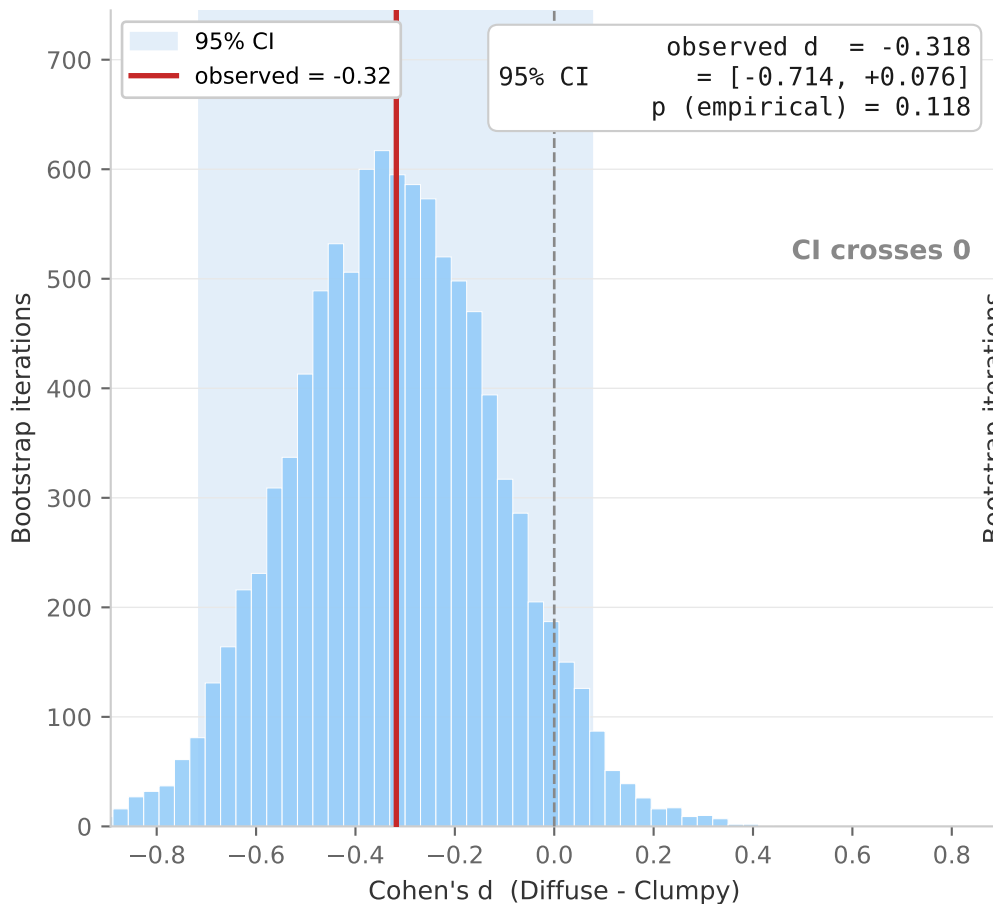
Its link to topic switching ($r = +0.16$) and time switching ($r = -0.18$) is essentially zero.

The manipulation reshapes writing rhythm specifically - not topic exploration or reading pace.

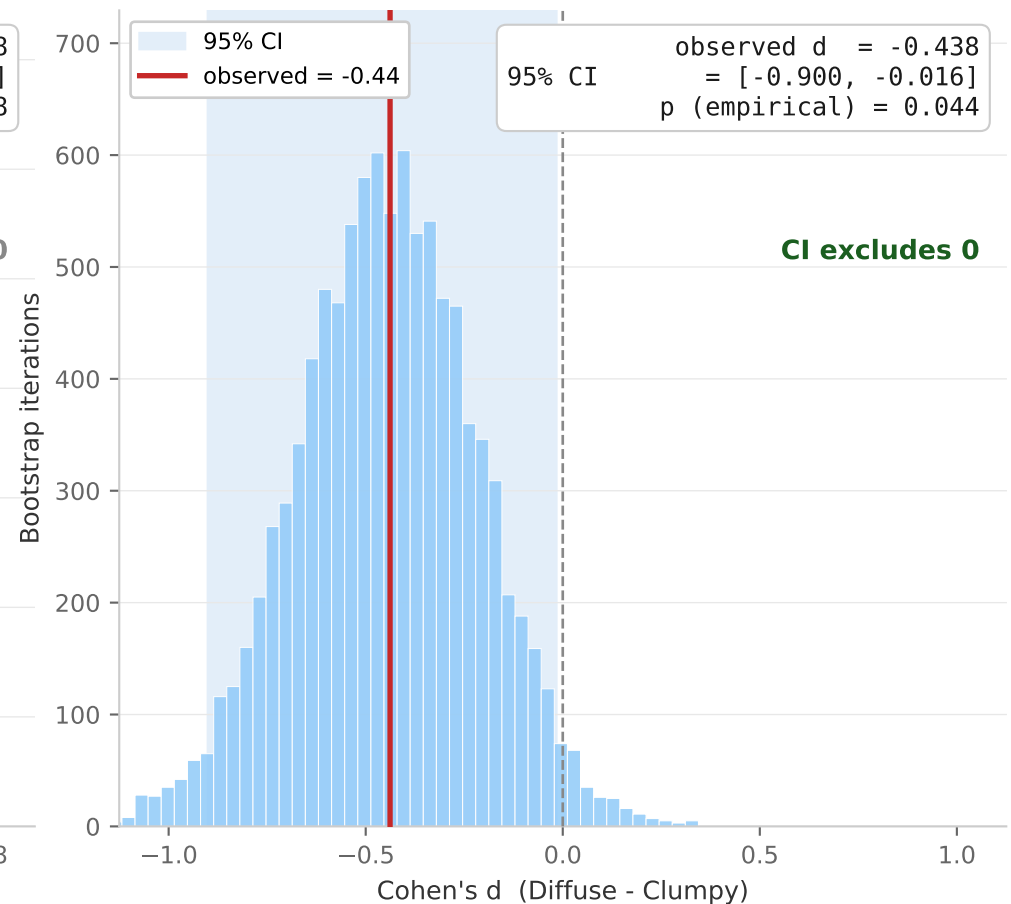
Bootstrap confidence intervals for the Mean reading-only streak effect

10,000 stratified resamples of participants (N = 99: 52 Diffuse, 47 Clumpy)

Raw Cohen's d (no covariate)



Adjusted Cohen's d (controls for answer length)



Each histogram = the Cohen's d we would observe across 10,000 resamples of the same N participants.

The raw effect's 95% CI crosses zero; after controlling for answer length, the CI excludes zero.

The Mean reading-only streak effect is robust to resampling once output volume is accounted for.